

Assignment 1 (Introduction)

1. Define Cloud Computing according to NIST? What are different essential characteristics of cloud according to NIST?
2. List out different applications of cloud computing? Explain any two in details?
3. What do you mean by NIST architecture? Which are the five cloud actors, draw the figure and explain any two.
4. According to NIST; how many Cloud Service Models are there? Explain any one in details?
5. List out different Cloud Deployment Models? Explain them
6. What are the factors that affect the Design principles of Cloud Architecture?
7. Define Infrastructural Constraints? What are different Infrastructural Constraints of Cloud Computing? Explain any two in details?
8. Write short notes on Evolution of Cloud Computing?
9. Discuss about the Jericho Cloud Cube Model? Explain with figure
10. Write short notes on:
 - i) Challenges of Cloud Computing
 - ii). Ethical Issues in Cloud Computing

Unit 2: Virtualization

1. Define Virtualization? Draw the figure of type 1 and type 2 virtualization?
2. Differentiate between type 1 and type 2 virtualization?
3. Define the following:
 - Virtual machine
 - Hypervisors
 - Containerization
 - Kubernetes
4. List out different Virtualization technologies? And explain any two of them.
5. Describe the relationship between Containerization and Kubernetes.
6. Write short notes on Virtual Machine Management and Configuration?
7. List out different Virtualization Level and explain them?
8. What do you mean by Virtual Private Cloud (VPC)?
9. How will you define Virtual Private Network (VPN) in Virtualization?
10. Write short notes on:
 - Multihoming
 - DNS service and cloud network

Assignment 3: Storage in Cloud Computing

1. Write short notes on DBaaS? What are the features, advantages and disadvantages of DBaaS?
2. What are different storage model of cloud computing? Explain them in your own words.
3. Differentiate between Parallel file systems and Distributed file system in cloud computing?
4. Differentiate between Relational and Non-relational Databases?
5. Differentiate between NoSQL and Graph Database?
6. What do you mean by elastic storage explain in short?
7. List out different cloud storage service providers in cloud computing?

Assignment 4: Resource Management in Cloud Computing

1. What do you mean by Resource Management in cloud computing?
2. What are the key points of Resource Management of cloud computing?
3. Define Scaling? What are different types of scaling in cloud computing? Explain it?
4. Differentiate between horizontal scaling and vertical scaling?
5. List out the advantages and disadvantages of scaling?
6. Write a short note about load balancing in cloud computing?
7. Name different types of Load Balancing? Explain any two?
8. What are the advantages and disadvantages of load balancing?
9. List out different components of Cloud Monitoring mechanism
10. How do you define Service Level Agreement and its types? What are common SLA Metrics?
11. What are different components of Cloud Billing Management?

Assignment 5 (Security in Cloud Computing)

1. Define Cloud Security? What are the Objectives of Cloud Security?
2. List some of the Cloud Security Threats?
3. Name different Security Mechanisms in Cloud Computing?
4. What are the benefits of Cloud Security?
5. Write short notes on Security as a Service (SECaaS)?
6. Make a table of Dimension of cloud security and write short description on it?
7. What are different Phases of Cloud Disaster Management?